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# Inflation versus Labor-Market Attention in FOMC Statements and Minutes

## A descriptive NLP report on the Federal Reserve dual mandate, 2000–2024

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### Abstract

This report compares how FOMC statements and minutes discuss the two sides of the Federal Reserve’s mandate: inflation and the labor market. The corpus contains 199 meetings from 2000 to 2024, with one statement and one set of minutes for each meeting. The main measure is a transparent dictionary balance: inflation mentions minus labor-market mentions, normalized by document length. Statements are shorter, more concentrated, and more inflation-heavy than minutes. Minutes are longer and closer to balance, with more space for labor-market conditions, financial stress, and uncertainty. I then audit the dictionary at the sentence level. A coverage-enriched sample of 3,000 sentences is labelled with a local language model and used to train simple classifiers based on TF-IDF and local embeddings. Conditional on these weak labels, both trained classifiers outperform the dictionary rule by a large margin, with TF-IDF slightly ahead of embeddings. The sentence model mainly reveals what the dictionary leaves out: financial-conditions language, especially in minutes.

### Introduction

The Federal Reserve has a dual mandate: price stability and maximum employment. After each FOMC meeting, the Committee publishes a short statement. A few weeks later, it publishes longer minutes. Both documents refer to the same meeting, but they serve different communication roles. The statement gives the immediate public message; the minutes give a fuller account of the discussion.

This report asks whether the two documents give the same relative attention to inflation and the labor market:

*Do statements and minutes give the same relative attention to inflation and the labor market? Do they discuss the two subjects in the same way?*

The question matters because the mandate combines two objectives, while the communication format may emphasize them differently. Price stability is closely tied to credibility, inflation expectations, and the transmission of monetary policy. Labor-market language can instead describe slack, wage pressure, resilience, or recession risk. The same meeting can therefore produce a short statement centered on inflation and longer minutes that give more room to employment and financial conditions.

The report takes a macro-topic angle: inflation versus the labor market. Mathieu’s companion report studies the related policy-tone angle: hawkish versus dovish communication. The two views overlap, but the mapping is imperfect. Inflation language can support a hawkish signal, while labor-market language can describe wage pressure or resilient demand under a restrictive stance.

The empirical strategy starts with a document-level dictionary balance and then checks it at several levels: term composition, sentence matches, low-rank TF-IDF projections, rate-decision groups,

policy periods, and election windows. I then use LLM-assisted weak labels to train sentence classifiers with TF-IDF and embedding features. These methods serve as checks on the same contrast: inflation language versus labor-market language.

## Related Work

A large literature studies FOMC communication through its effect on financial markets, especially fixed-income instruments and Overnight Indexed Swap rates, which closely track market expectations of monetary policy. Tadle (Tadle, 2022) builds sentiment indices for statements and minutes and relates minutes sentiment to financial-market reactions. This report stays on the textual side and does not estimate market responses.

Several papers use text methods to study FOMC communication directly. Acosta and Meade (Acosta and Meade, 2015) study post-meeting statements with semantic similarity methods and document how statements changed over time. Hansen, McMahon, and Prat (Hansen et al., 2018) apply computational linguistics to internal FOMC transcripts to study deliberation and transparency inside the Committee. Their data are richer than mine because transcripts contain individual interventions. This report uses public statements and minutes, so it measures released communication rather than internal debate.

The closest paper for the dual-mandate motivation is Shapiro and Wilson (Shapiro and Wilson, 2022). They estimate central-bank objectives from the language of internal meetings, including the relative weight placed on inflation and real activity. My report is narrower: it describes how inflation and labor-market topics appear in public communication, without estimating the economic weight of each objective.

Doh, Kim, and Yang (Doh et al., 2021) show that qualitative descriptions in FOMC statements can matter for bond prices. Doh, Song, and Yang (Doh et al., forthcoming) use alternative FOMC statements and language models to measure hawkish and dovish policy stance. Those papers are closer to the companion report on policy tone.

## Data

The corpus comes from the official Federal Reserve website and is retrieved and parsed automatically. Each row corresponds to one meeting and contains the meeting date, the statement URL, the minutes URL, the statement text, and the minutes text.

The sample contains 199 meetings from February 2000 to December 2024. Every meeting has both a statement and minutes. Rate decisions are inferred from statement text with simple regular expressions and coded as *hike*, *cut*, or *hold*. The sample has 59 hikes, 30 cuts, and 110 holds.

Table 1: Corpus size and document length.

| Document type | Documents | Mean words | Median words | Std. words |
|---------------|-----------|------------|--------------|------------|
| Statements    | 199       | 526.1      | 536.0        | 217.4      |
| Minutes       | 199       | 7283.2     | 7536.0       | 2441.3     |

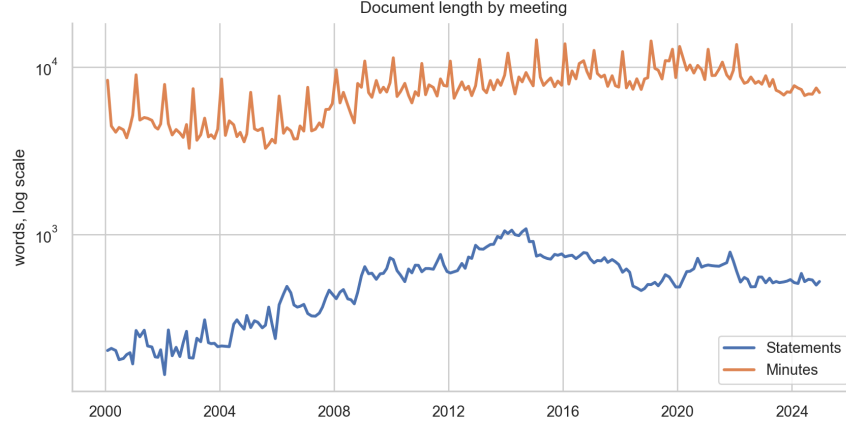


Figure 1: Document length by meeting. Minutes are much longer than statements, so dictionary scores are normalized by document length.

Minutes contain about fourteen times more words than statements on average. All dictionary scores are therefore reported as mentions per 1,000 words. Preprocessing is deliberately minimal: texts are lowercased, whitespace is normalized, and words are counted with a regular expression. Stopwords are kept because several dictionary entries are multi-word expressions, such as *price stability*, *labor market*, and *inflation expectations*.

### Dictionary balance

I define two short dictionaries. The inflation dictionary contains terms such as *inflation*, *inflationary*, *price stability*, *price pressures*, *inflation expectations*, *core inflation*, *PCE inflation*, *consumer prices*, *energy prices*, *food prices*, *cost pressures*, *supply constraints*, and *disinflation*.

The labor-market dictionary contains terms such as *employment*, *unemployment*, *labor market*, *job gains*, *payrolls*, *hiring*, *layoffs*, *wages*, *labor demand*, *labor supply*, *labor force*, *slack*, *participation*, *vacancies*, and *job openings*.

For a document  $d$ , the two attention scores are:

$$I(d) = 1000 \times \frac{\#\{\text{inflation terms in } d\}}{\#\{\text{words in } d\}}, \quad L(d) = 1000 \times \frac{\#\{\text{labor-market terms in } d\}}{\#\{\text{words in } d\}}. \quad (1)$$

The balance score is:

$$B(d) = I(d) - L(d). \quad (2)$$

A positive value means that inflation terms are more frequent than labor-market terms. The score is a coarse measure of relative textual attention. Its limits are mechanical: phrases can overlap with single words, and words such as *wages* can appear in labor-market descriptions as well as inflation-pressure narratives. Because this score is the main measure, I audit it through term counts and sentence-level matches.

The sentence check tags each sentence as *inflation only*, *labor only*, *both*, or *neither*, depending on exact dictionary matches. I also build a low-rank TF-IDF representation of statements and minutes. After vectorizing documents with unigrams and bigrams, I apply truncated singular value decomposition  $X = U_k \Sigma_k V_k^T$ . The first components capture the largest directions of variation in the TF-IDF matrix. If statements and minutes use systematically different vocabularies, formulas, or recurring expressions, this format effect should appear in the first dimensions.

## Document-Level Results

Inflation language is more frequent than labor-market language in both document types, and the gap is larger in statements. This fits the role of the statement as a short policy message. Minutes cover more background discussion, so the same macro topics take up a smaller share of the text. Labor-market vocabulary also becomes more frequent over the sample in both document types.

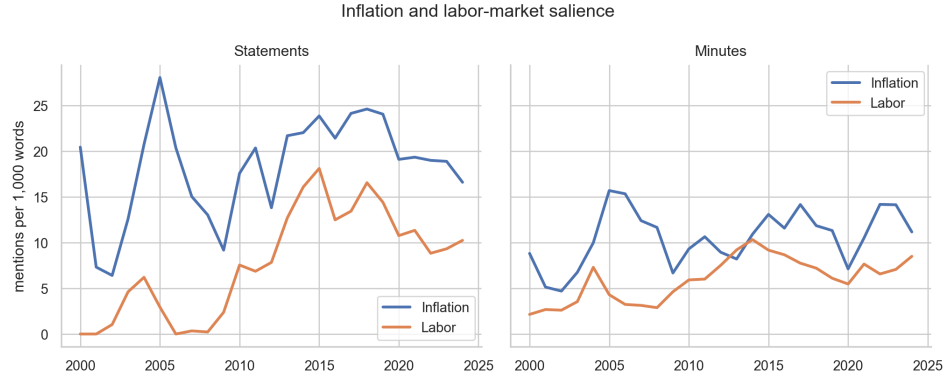


Figure 2: Yearly average inflation and labor-market attention. Statements are more concentrated than minutes on both dimensions.

Table 2: Average mention scores per 1,000 words.

| Document type | Inflation | Labor market | Balance |
|---------------|-----------|--------------|---------|
| Statements    | 18.37     | 7.75         | 10.62   |
| Minutes       | 10.58     | 5.98         | 4.60    |

The balance is usually positive, but the statement and minutes series move differently. Statements are often more inflation-dominant than minutes. The gap narrows when labor-market language becomes more visible, especially around the first Covid shock. This is one of the clearest cases where the labor side of the mandate appears directly in the statement.

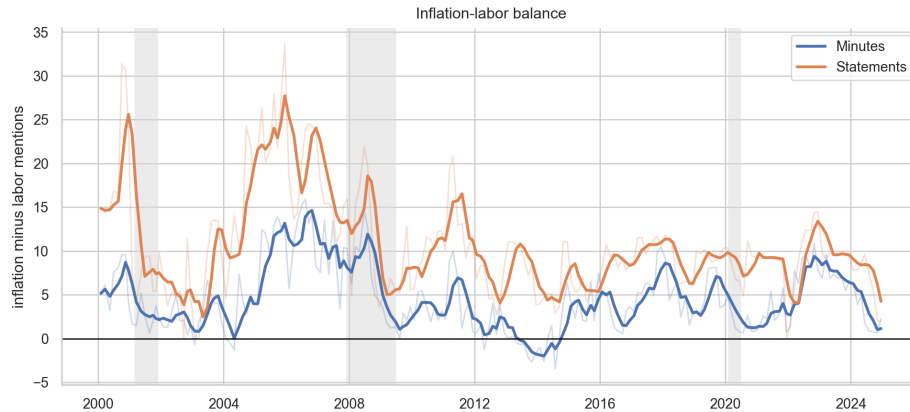


Figure 3: Inflation-minus-labor balance by meeting. Faint lines show raw meeting scores; darker lines show a four-meeting rolling average.

## Term and sentence checks

Table 3: Most frequent dictionary terms in the corpus. Counts use the same matching rule as the score.

| Document type | Topic     | Most frequent terms                                                                                                  |
|---------------|-----------|----------------------------------------------------------------------------------------------------------------------|
| Statements    | Inflation | inflation (1,340); inflation expectations (267); price stability (259); energy prices (76); core inflation (22)      |
| Statements    | Labor     | labor market (410); employment (337); unemployment (138); job gains (64); slack (18)                                 |
| Minutes       | Inflation | inflation (10,387); inflation expectations (1,587); energy prices (985); price stability (779); core inflation (410) |
| Minutes       | Labor     | employment (2,293); labor market (2,235); unemployment (1,953); participation (444); slack (442)                     |

The term audit shows that the inflation score is driven mainly by *inflation*, but also by coherent expressions such as *inflation expectations*, *price stability*, and *core inflation*. The labor score is spread across *labor market*, *employment*, and *unemployment*, with additional contributions from *job gains*, *participation*, and *slack*.

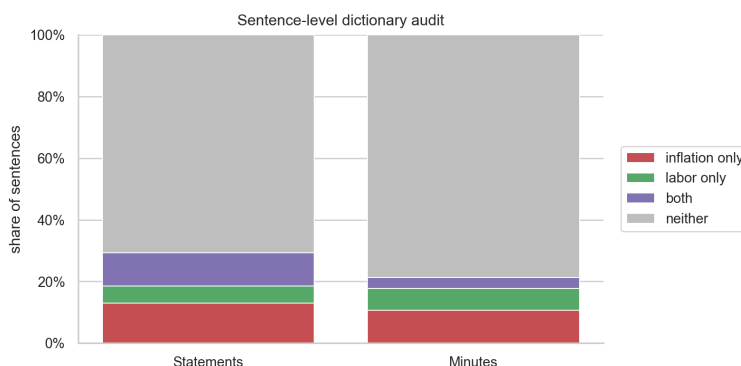


Figure 4: Sentence-level dictionary check. Each sentence is classified as inflation only, labor only, both, or neither.

The sentence check gives the same document-format picture. Statements often compress both sides of the mandate into a short text. Minutes spread the discussion across many more sentences, including many sentences outside the two dictionaries.

## Decisions, periods, and document gaps

Hike meetings have higher average balances than cuts or holds. The association is plausible: meetings near hikes use more inflation language relative to labor-market language. The points should be read with care because neighboring meetings are serially correlated, which creates visible clusters in the figure.

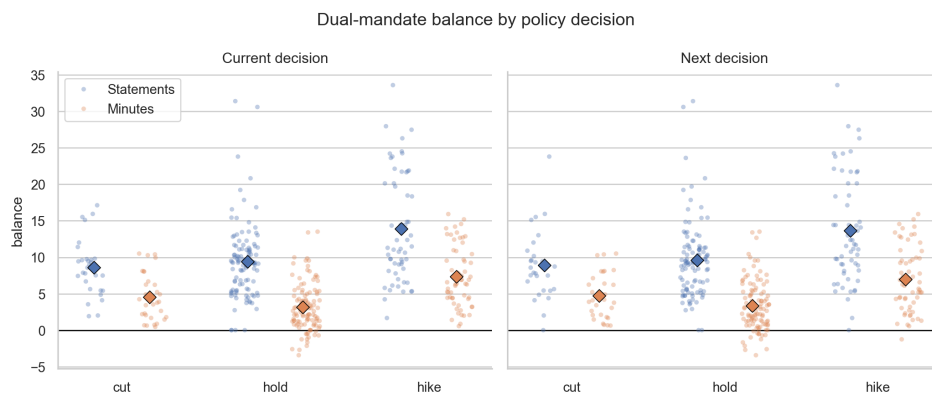


Figure 5: Mandate balance by current and next rate decision. Points are meetings; diamonds are group means.

The period comparison locates the main episodes. The Covid shock has a lower balance, mainly because labor-market language becomes more visible in statements. The global financial crisis remains more positive in this dictionary, probably because much of the discussion uses financial-stability, credit, and risk language outside the labor terms.

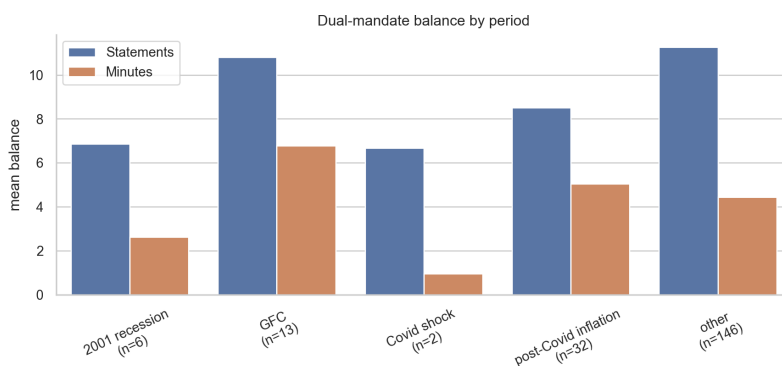


Figure 6: Average mandate balance by period. Sample sizes are shown under the period labels.

The statement–minutes gap shows how much the immediate policy message differs from the longer account of the same meeting. Large negative gaps correspond to meetings where the statement is more inflation-heavy than the minutes.

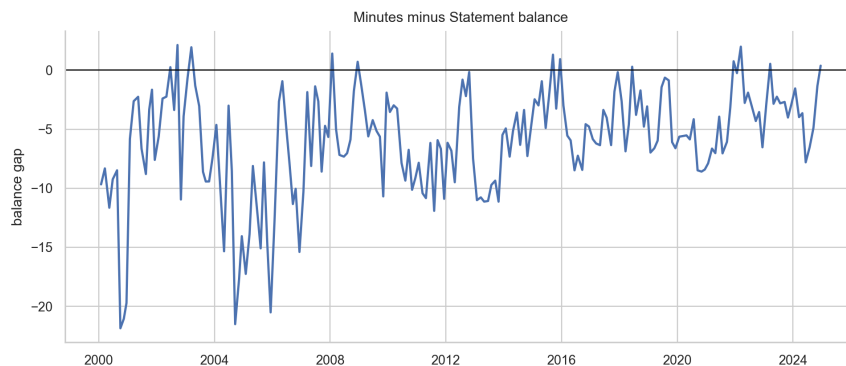


Figure 7: Minutes minus statement balance. Negative values mean that statements are more inflation-dominant than minutes.

### Low-rank TF-IDF projection

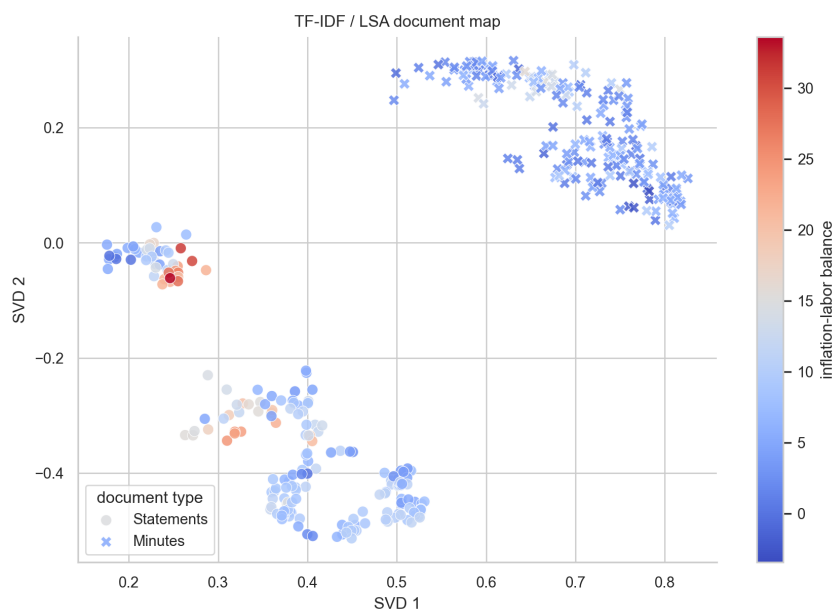


Figure 8: TF-IDF/LSA map of statements and minutes. Color is the inflation-labor balance.

The first two SVD dimensions separate statements and minutes clearly. In this low-rank TF-IDF map, document format is a major source of textual variation. Statements also split into two visible clusters, one of them with especially high inflation-labor balance values. This suggests that some statement templates are disproportionately inflation-heavy.

### Sentence-Level Weak Labels and Topic Classification

The dictionary score is transparent, but narrow. Many important FOMC sentences use neither inflation nor labor-market vocabulary. During stress periods, for example, communication often turns to credit, market functioning, liquidity, asset prices, or financial stress. These sentences can be central to the meeting narrative while being classified as *neither* by the inflation-labor dictionary.

I therefore add a broader sentence-level audit. The task asks whether a sentence can be assigned to one of five broad FOMC topics: inflation, labor, financial conditions, policy decision, or other. This

gives a supervised NLP component while keeping the dictionary balance as the main interpretable document measure.

### Sentence table and weak annotation

I split all statements and minutes into sentences with a punctuation-based rule and drop very short sentences. This gives 58,976 usable sentences. For each sentence, I keep the meeting date, document type, sentence position, rate decision, period, calendar indicators, document-level dictionary balance, and sentence-level dictionary label.

I then construct a coverage-enriched sample of 3,000 sentences for annotation. The sample covers statements and minutes, the four dictionary sentence buckets, meetings with large statement–minutes gaps, crisis periods, the post-Covid inflation period, and federal election windows. Remaining slots are filled with additional sampled sentences. This sampling design is useful for auditing the dictionary, but raw label shares in the annotated sample should not be interpreted as corpus frequencies.

The 3,000 sentences are labelled with a local language model, *llama3.2*, using a fixed JSON schema. The main label is *main\_topic*, with six possible values: *inflation*, *labor*, *both*, *financial\_conditions*, *policy\_decision*, and *other*. The prompt asks the model to classify only from the sentence itself. Sentences about credit, markets, liquidity, volatility, asset prices, or market functioning are assigned to *financial\_conditions*.

The annotations are weak labels generated under a fixed topic schema. The classifier results below measure agreement with this schema rather than accuracy against a manually validated benchmark.

Dictionary labels versus weak LLM sentence labels

| Dictionary sentence label | inflation | labor | both | financial_conditions | policy_decision | other |
|---------------------------|-----------|-------|------|----------------------|-----------------|-------|
| inflation                 | 250       | 1     | 22   | 131                  | 1               | 7     |
| labor                     | 1         | 137   | 91   | 41                   | 1               | 20    |
| both                      | 30        | 1     | 103  | 62                   | 1               | 9     |
| other                     | 14        | 19    | 48   | 1116                 | 163             | 731   |
| LLM main topic            |           |       |      |                      |                 |       |
|                           |           |       |      |                      |                 |       |

Figure 9: Dictionary sentence labels compared with LLM-assisted sentence topics. The main gap is that many dictionary-*other* sentences are labelled as financial conditions.

The comparison between dictionary labels and LLM-assisted labels identifies the main blind spot of the dictionary. Many sentences with inflation terms are labelled as inflation or both, and many sentences with labor-market terms are labelled as labor or both. The main disagreement comes from dictionary-*other* sentences labelled as *financial\_conditions*. Financial-stress language often matters for the meeting narrative without using either inflation or labor-market terms.

### Sentence classifiers

The supervised input is one sentence. The target is the LLM-assisted *main\_topic* label. For evaluation, I merge *policy\_decision* and *other* into a single *other\_or\_policy* class, because both are non-mandate categories and the smaller class would make the comparison noisier. The final target classes are therefore: *inflation*, *labor*, *both*, *financial\_conditions*, and *other\_or\_policy*.

I compare four ways of assigning a topic to a sentence:



1. The *dictionary rule* maps the sentence-level dictionary label directly into the supervised classes. It has no training step and serves as the transparent baseline.
2. The *embedding prototype* baseline embeds short descriptions of each topic and assigns each sentence to the nearest topic description. It uses embeddings, but no supervised training.
3. The TF-IDF model represents each sentence with unigram and bigram TF-IDF features and trains a balanced logistic regression.
4. The embedding model represents each sentence with a local `nomic-embed-text` vector and trains the same balanced logistic regression on dense features.

The two logistic-regression models are evaluated with stratified cross-validation on the weakly labelled sample. Macro-F1 is the main metric because the classes are unevenly distributed and minority topics matter for interpretation.

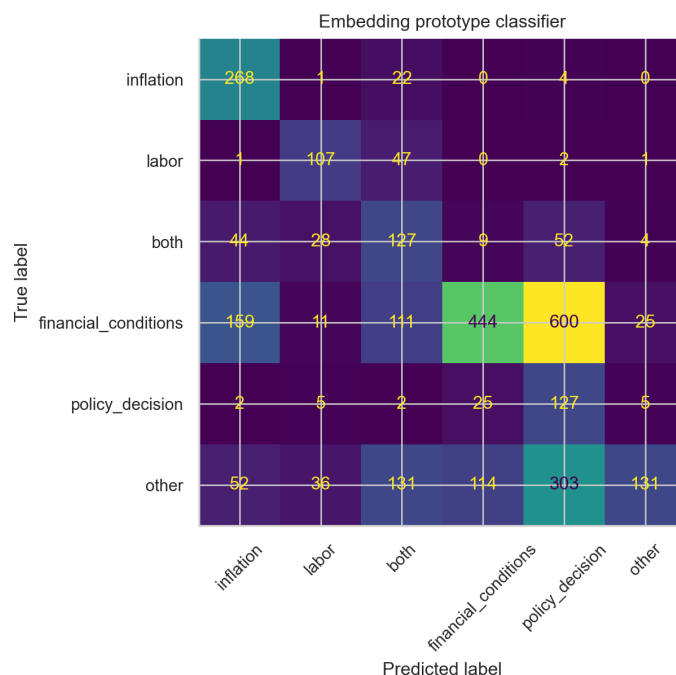


Figure 10: Embedding prototype baseline on the LLM-assisted sentence sample. This model has no training step: each sentence is assigned to the nearest embedded topic description.

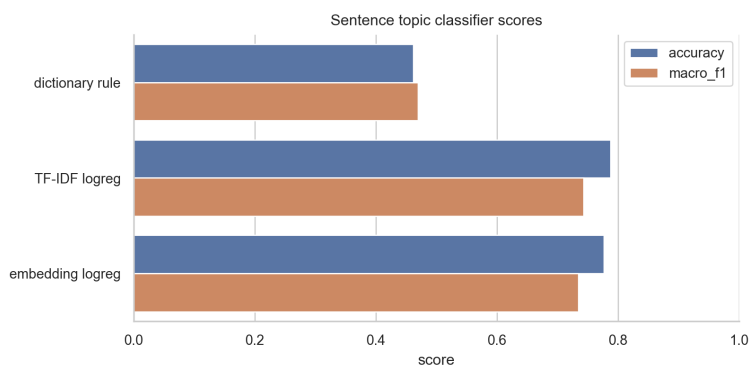


Figure 11: Sentence-level classifier scores. Macro-F1 is the main metric because the weak labels are unevenly distributed.

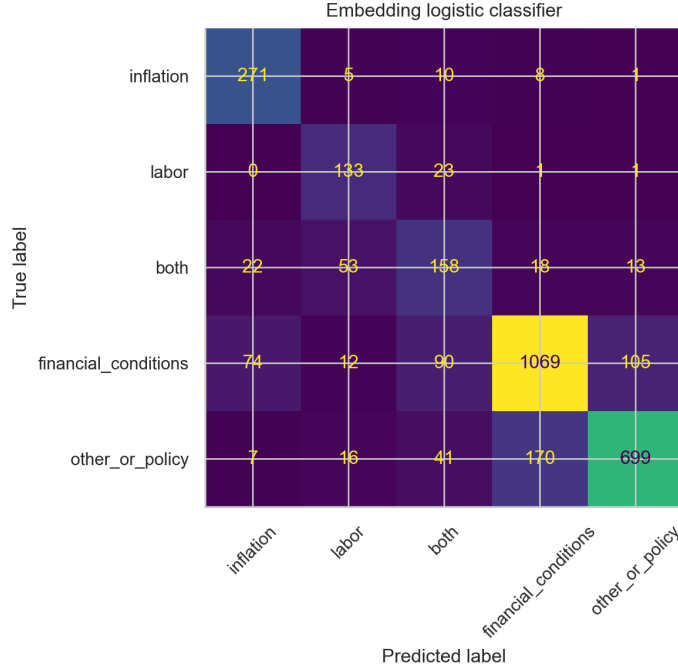


Figure 12: Confusion matrix for the embedding logistic-regression sentence classifier.

Table 4: Sentence-topic classification against LLM-assisted weak labels.

| Model                           | Accuracy | Macro-F1 |
|---------------------------------|----------|----------|
| Dictionary rule                 | 0.461    | 0.469    |
| Embedding prototype             | 0.401    | 0.429    |
| TF-IDF + logistic regression    | 0.787    | 0.743    |
| Embedding + logistic regression | 0.777    | 0.734    |

Once the weak-label schema is fixed, both supervised models recover the assigned sentence topics much better than the dictionary rule and the no-training prototype baseline. The TF-IDF model is slightly ahead of the embedding model. This fits the corpus: many FOMC topics have stable phrases, and sparse unigram-bigram features capture them well. The embedding model remains useful as a compact semantic representation, especially when related sentences use different surface words.

### Full-corpus sentence topics

After cross-validation, I fit a sentence-topic classifier on the annotated sample and apply it to all 58,976 sentences. In the notebook run used for the figures, full sentence embeddings are available, so the full-corpus predictions use the embedding logistic-regression model. If full embeddings are unavailable, the notebook falls back to the TF-IDF classifier. The resulting file `data/processed/fomc_sentence_topic_predictions.csv` contains one row per sentence, with predicted topic, confidence, meeting date, document type, period, and calendar variables.

The cross-validation table evaluates held-out agreement with the weak labels. The full-corpus prediction uses the fitted classifier as a descriptive measurement device. The most reliable comparisons are relative: statements versus minutes, crisis versus non-crisis periods, and dictionary balance versus classifier balance.

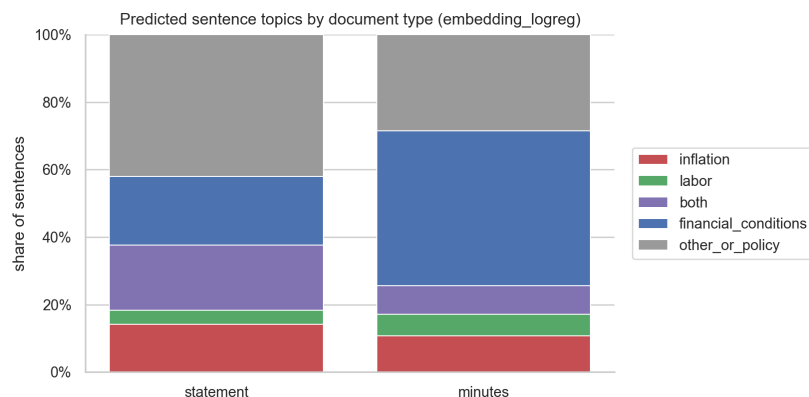


Figure 13: Predicted sentence-topic shares for statements and minutes. The classifier view keeps the inflation–labor contrast but adds financial conditions and policy-decision language.

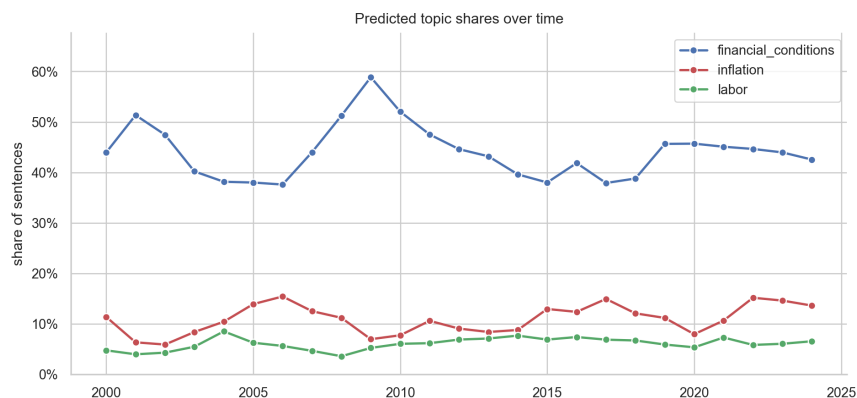


Figure 14: Predicted sentence-topic shares over time. See peak around 2008 for financial condition.

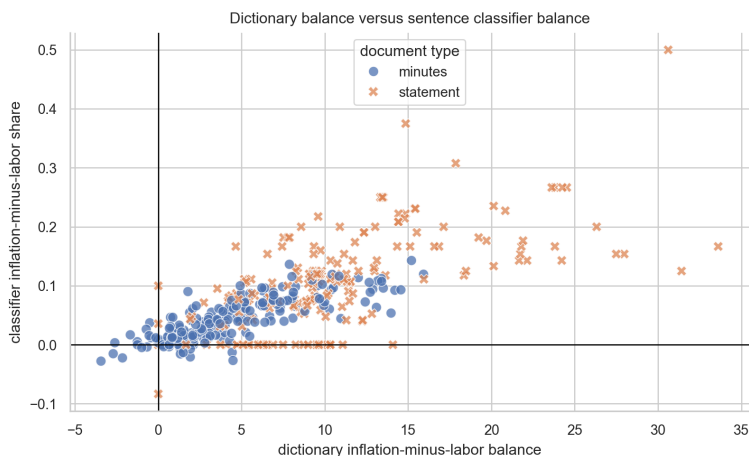


Figure 15: Dictionary inflation-minus-labor balance against the classifier inflation-minus-labor sentence share. Each point is a meeting-document pair.

The classifier assigns a larger financial-conditions share to minutes than to statements, and a larger combined-mandate share to statements than to minutes. This matches the format interpretation from

the dictionary and SVD checks. Statements compress the public message into a short text; minutes contain more background discussion, including financial conditions.

The scatterplot links the two approaches. The dictionary balance measures exact inflation terms minus exact labor-market terms per 1,000 words. The classifier balance measures the predicted share of inflation sentences minus the predicted share of labor sentences. The two measures broadly move together. Their disagreements help identify documents where financial-conditions or policy-decision sentences occupy space that the dictionary leaves outside the mandate balance.

### Election-window check

I add a calendar check for presidential election years, midterm years, meetings within 90 days of a federal election, first meetings of the year, even-year Board-cycle years, and Chair appointment or reappointment years. These variables are descriptive markers for scheduled institutional and electoral dates.

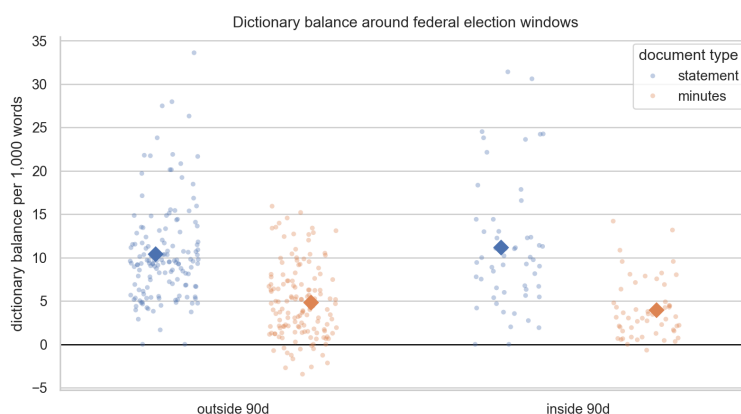


Figure 16: Dictionary balance inside and outside 90-day federal election windows. Points are meetings; diamonds are group means.

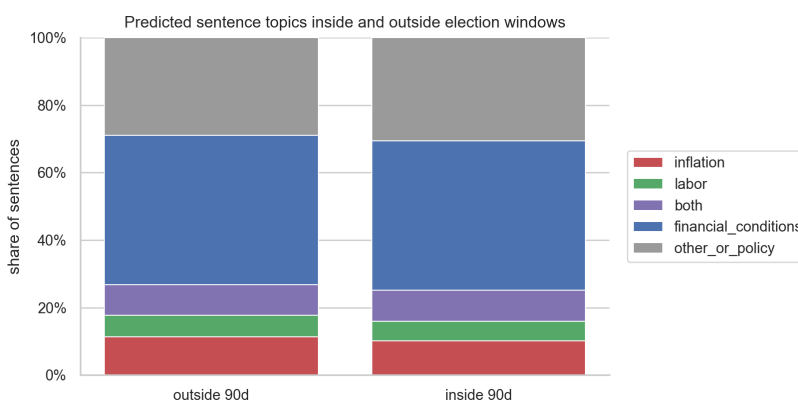


Figure 17: Predicted sentence-topic shares inside and outside 90-day federal election windows.

A visible shift in topic mix around these dates would be worth documenting. The comparison is coarse: the election window is mechanical, the sample is small, and the timing does not identify political influence. In this sample, the differences are negligible.

## Discussion

The main result is a document-format effect. Statements are short and more inflation-heavy. Minutes are longer, more diffuse, and closer to balance. The statement–minutes gap shows how much the immediate policy message differs from the longer meeting account.

The checks point in the same direction. The dictionary score is higher in statements. The term audit shows that the counts are driven by economically meaningful expressions. The sentence-level dictionary audit shows how statements compress mandate language. The low-rank TF–IDF map shows that statements and minutes occupy different textual regions. The weak-label classifier adds one more piece: broader topics, especially financial conditions, can be recovered from sentence text even when exact mandate terms are absent. The convergence of these checks makes a pure dictionary artifact less likely.

The pattern is consistent with the communication role of statements. A short policy statement has to foreground the rationale for the decision, and inflation is closely tied to credibility, expectations, and the transmission of monetary policy. Minutes include more of the discussion around risks, uncertainty, credit, market functioning, and labor conditions.

The sentence-level results also clarify the limits of a pure dual-mandate dictionary. In stress periods, many important sentences are about financial conditions, liquidity, and market functioning. These sentences belong to FOMC communication, but they are outside the narrow inflation–labor contrast. The classifier view therefore adds a third dimension that explains part of what the dictionary leaves out.

Rate-decision and period comparisons move in plausible directions. Hike meetings are more inflation-heavy than cut meetings, and the Covid shock is one of the clearest cases where labor-market language becomes more visible. Given the small sample and serial dependence between meetings, these comparisons should be read as descriptive associations. The election-window check is also descriptive and mostly negative: topic mix inside and outside 90-day federal election windows looks very similar.

The implication is that document choice matters. A study based only on statements will mechanically emphasize a different part of the mandate than a study based on minutes. Models that pool both formats can easily learn format differences before learning a substantive inflation–labor contrast.

## Conclusion

In this sample of 199 FOMC meetings, statements and minutes give different textual weights to the two sides of the dual mandate. Statements are much shorter and more inflation-dominant. Minutes are more balanced and contain a wider discussion. This is the main descriptive result of the report.

Several checks support this result. The dictionary balance is higher in statements. The term audit shows that the score is driven by meaningful monetary-policy expressions, although the word *inflation* plays a large role. The sentence-level dictionary audit shows that statements compress inflation and labor-market language into a short document. The low-rank TF–IDF map shows that statements and minutes are different textual formats. The weak-label sentence classifier shows that broader sentence topics are learnable from text, with TF–IDF and embedding features far ahead of the dictionary rule on the weak-label benchmark.

The inflation–labor contrast captures only part of FOMC communication. In stress periods, financial conditions become central, and a pure dual-mandate dictionary misses that part of the story. The classifier view makes this visible, especially in minutes. Its absolute topic shares should be read cautiously because the model is trained on a small weakly labelled sample, but its relative comparisons help interpret where the dictionary balance succeeds and where it is incomplete.

These results are descriptive NLP evidence, not an assessment of the Federal Reserve’s true mandate weights. The measures developed here could be used in a broader study of monetary-policy communication, but they would need to be combined with policy decisions, macroeconomic conditions, and market reactions before supporting claims about actual policy preferences.

## A TF-IDF Sanity Check



Figure 18: Macro-F1 scores for TF-IDF models. The dashed line is the majority-class baseline.

A small TF-IDF classification exercise remains in the appendix. The target is either the current rate decision or the next meeting's rate decision. The models are standard TF-IDF+logistic regression and TF-IDF+linear SVM classifiers, compared with a majority-class baseline. The split is chronological: the first 70% of meetings are used for training and the last 30% for testing.

The rate-decision scores remain modest. The majority baseline has macro-F1 0.222. For the current decision, the best model is TF-IDF+linear SVM on full minutes, with macro-F1 0.418. For the next decision, the best model is again based on minutes, with macro-F1 0.396. This appendix check suggests that minutes contain more decision-related context than statements, even though statements have a higher density of mandate terms.

## B Document-Level Embedding Check

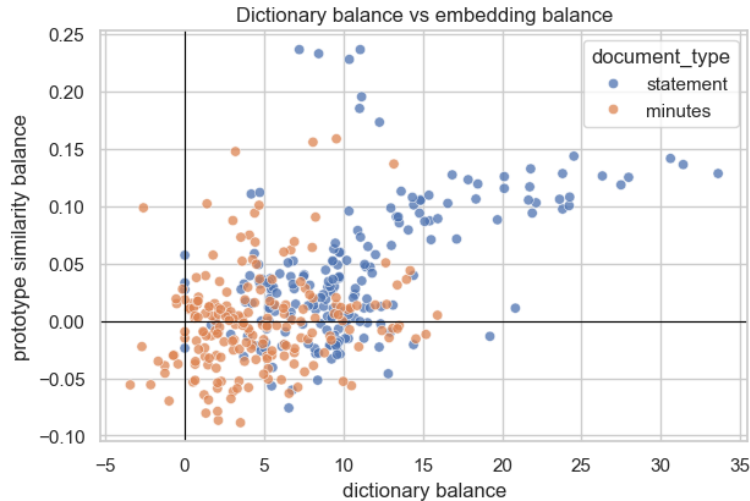


Figure 19: Dictionary balance against embedding prototype balance.

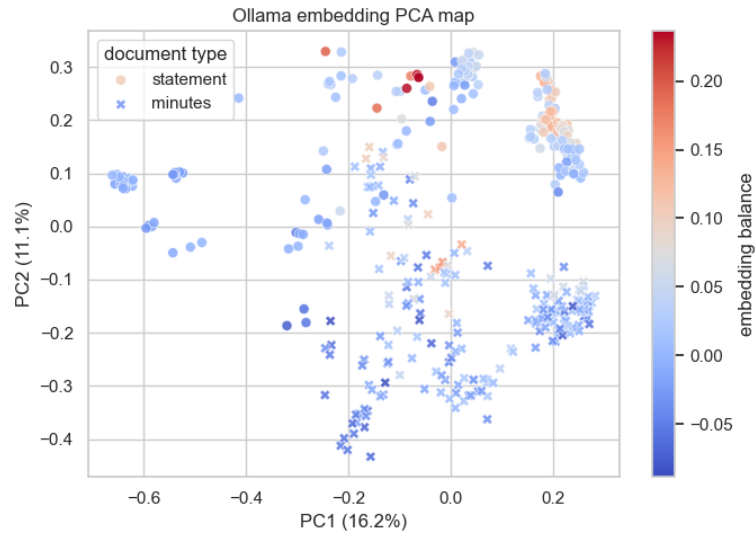


Figure 20: PCA map of local document embeddings. Color is the embedding prototype balance.

Before the sentence-level model, I also tried a local document-level embedding check with `nomic-embed-text`. Each document is represented by a short beginning/middle/end excerpt, and the document vector is compared with one inflation prototype phrase and one labor-market prototype phrase. This document-level view is noisier than the sentence classifier, especially for minutes, so I keep it in the appendix. It points to the same message as the LSA map: statements and minutes are different document formats.

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